

Kangning (Ken) Huang

Assistant Professor of Environmental Studies

New York University Shanghai

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EMPLOYMENT

- 2022.9–present** Assistant Professor of Environmental Studies, NYU Shanghai
2022.1–2022.8 Visiting Assistant Professor Faculty Fellow, NYU New York
2020.1–2021.12 ASP Postdoctoral Fellow, National Center for Atmospheric Research (NCAR)

EDUCATION

- 2020** Ph.D., Forestry and Environmental Studies, Yale University, School of the Environment
2014 M.S., Cartography and Geographic Information System, Sun Yat-sen University
2011 B.S., Geographic Information System, Sun Yat-sen University

RESEARCH INTERESTS

Environmental impact from urbanization and climate change; Future urbanization and its interaction with climate change; Urban climate impact assessment and reduction.

AWARDS

- 2020** NCAR Advanced Study Program Postdoctoral Fellowship
2017 NASA Earth and Space Science Fellowship
2012 P. R. China National Graduate Scholarship

GRANTS

Active / Completed

- 2026–2028** “Boom or bust: accessibility tipping point underlying divergent urban transitions in China.” NYU Arts & Science Fund for Research on China (\$50,000). Co-PI with Mingzhen Lu.
2024–2026 “Mapping the Mismatch: The Urbanization of the Built Environment Versus People in Modern China.” NYU Arts & Science Fund for Research on China (\$50,000).
2022–2025 “Maximizing Urban Climate Under Resource Constraints: An International, Interdisciplinary Approach.” Shanghai Leading Talents Program, Distinguished Young Scholars, Shanghai Municipal Government (~\$100,000).
2022–2023 “Future Heat Extremes in China’s Urban Slums: Projections and Mitigations.” Google Award for Inclusion Research program (\$60,000).
2022–2023 “Climate trade-offs in future urban growth: should cities sprawl or density?” NYU Climate Change Initiative Seed Grants (\$10,000).
2017–2019 “Modeling the water requirement for urban heat island mitigation with multi-sensor and multi-temporal remote sensing data.” NASA Earth and Space Science Fellowship (NESSF) Program (\$120,000).
2017–2018 “The compound heat extreme in China from the urban heat island effect and climate change.” Yale Tropical Resources Institute Fellowship, [Yale Hixon Fellowship Grant](#), and Yale Institute for Biospheric Studies Doctoral Pilot Grant (\$16,400).

Declined

- 2025** “Integrating blue and green infrastructure to promote urban climate resilience in compound drought-heatwave.” National Science Foundation of China.
- 2024** “Digital intervention for mental health in compact, low-carbon urban environments.” China Medical Board.
- 2023** “The water limits to climate adaptation by urban greening.” Amazon Research Award.

PUBLICATIONS

Articles, Under Review / Preprint

1. Yangzi Che, Guangdong Li, Xiaoping Liu, Libang Ma, **Kangning Huang**. “Stage-dependent scaling of travel distance shaped by 3D urban form.” Submitted to *PNAS*.
2. **Kangning Huang**, Aiyu Zheng, Hannes De Deurwaerder, Brian Stone Jr., Mingzhen Lu. “Urban vine canopies as a heat-adaptation solution.” In preparation for *Nature Climate Change*.
3. Yimin Chen, Jing Qian, Yijing Wang, Xiaoping Liu, Ming Luo, Yutian Liang, Weilin Liao, Zhu Deng, **Kangning Huang**. “Green AI potential stranded by geopolitical misalignment.” Submitted to *Nature*.
4. Meizi You, ChengHe Guan, Yuming Guo, Bao-Jie He, Steven Jige Quan, John S. Ji, Jianxiang Huang, Rongbin Xu, Zhihu Xu, Tianren Yang, Longxu Yan, **Kangning Huang**, Chengwei Li, Wei Zhai, Chaosu Li, Wenhua Yu, Tingting Ye, Xiyuan Ren, Yanming Liu, Zhengyu Yang, Waishan Qiu. “Global multi-city heat perception: seasonal dynamics and climate–health signals.” Under review in *Nature Communications*.
5. **Kangning Huang**, Mingzhen Lu. “Nested economies of scale in global city mass.” Under review in *Nature Cities*.
6. **Kangning Huang**, Micheal Staudenmaier, Olga Wilhelmi. “Rising heat risks in the US by the mid-21st century.” Under review in *Climate*.
7. Jiayong Liang, Imman Hilaly, Xuyan Gao, ChengHe Guan, Ying Li, **Kangning Huang**. “Integrating building height and protection standards in global urban flood risk assessment.” Under review in *Scientific Reports*.
8. Yupeng Liu, **Kangning Huang**, Shengping Li, Karen C. Seto, Wei-Qiang Chen. “Material-saving strategies for decarbonizing the global building stock.” Submitted to *Nature Geoscience*.
9. Xucai Zhang, Xiaobing Wei, Xiaoping Liu, Honghong Wei, Yue Zheng, Hongkai Gu, Yangzi Che, Yeran Sun, Ming Luo, Weilin Liao, Haosheng Huang, **Kangning Huang**. “Toward Cooler Cities by Homogeneous Scale Planning.” In revision at *PNAS*.

Peer-Reviewed Articles

([Google Scholar](#) citations: 4,634; h-index: 22)

1. Yujie Sun, Xuyan Gao, Jiayong Liang, **Kangning Huang**. 2026. “Unveiling the causal link between informal settlement demolition and urban cooling.” *npj Environmental Social Sciences*. 1, 7.
2. **Kangning Huang**, ChengHe Guan, Brian Stone Jr., Jiayong Liang. 2025. “Declining urban density attenuates rising population-weighted exposure to surface heat extremes.” *Scientific Reports*. 15(1), 13860.
3. Jiayong Liang, Desheng Liu, Lihan Feng, **Kangning Huang**. 2025. “Rapid Probabilistic Inundation Mapping Using Local Thresholds and Sentinel-1 SAR Data on Google Earth Engine.” *Remote Sensing*. 17(10), 1747.
4. Ren, Xiyuan, ChengHe Guan, Shengze Chen, Meizi You, Ying Li, and **Kangning Huang**. 2025. “Planning for Rhythimized Urban Parks: Temporal Park Classification and Modes of Action.” *Journal of the American Planning Association*. 1–20.

5. Shasha Wang, Wenfeng Zhan, Bingbing Zhou, Shilu Tong, TC Chakraborty, Zhihua Wang, **Kangning Huang**, Huilin Du, Ariane Middel, Jiufeng Li, Zihan Liu, Long Li, Fan Huang, and Manchun Li. 2025. “Dual impact of global urban overheating on mortality.” *Nature Climate Change*. 15, 497–504.
6. Zhaowu Yu, Siheng Li, Wenjun Yang, Jiquan Chen, Mohammad A Rahman, Chenghao Wang, Wenjuan Ma, Xihan Yao, Junqi Xiong, Chi Xu, Yuyu Zhou, Jike Chen, **Kangning Huang**, Xiaojiang Gao, Rasmus Fensholt, Qihao Weng, Weiqi Zhou. 2025. “Enhancing Climate-Driven Urban Tree Cooling with Targeted Non-climatic Interventions.” *Environmental Science & Technology*. 59(18), 9082–9092.
7. Pei, Yaolin, Xiang Qi, Gen Li, Weiming Tang, **Kangning Huang**, Brian J. Hall, and Bei Wu. 2025. “Unequal Effects of the Lockdown on Mental Health in Shanghai.” *Journal of Community Psychology*. 53(1): e23177.
8. Zhong, Xinyuan, Tingting Guo, Jianghui Zhang, Qiong Wang, Rong Yin, Kunpeng Wu, Qing Zou, Meng Zheng, Brian J. Hall, Andre M. N. Renzaho, **Kangning Huang**, Wen Chen. 2025. “Short-Term Effect of Air Pollution on Daily Hospital Visits for Anxiety Disorders in Southern China.” *Toxics*. 13(1): 45.
9. Jiayong Liang, Maria Adele Carrai, Ammar A. Malik, Xi Gao, Xuyan Gao, Lihan Feng & **Kangning Huang**. 2024. “Measuring rising heat and flood risk along the belt-and-road initiative.” *Discover Environment*. 2, 119.
10. Xu, Wenhao, Gongqin Wang, Shaoda Liu, Junfeng Wang, William H. McDowell, **Kangning Huang**, Peter A. Raymond, Zhifeng Yang, and Xinghui Xia. 2024. “Globally elevated greenhouse gas emissions from polluted urban rivers.” *Nature Sustainability*. 7, 938–948.
11. Huilin Du, Wenfeng Zhan, Bingbing Zhou, Yang Ju, Zihan Liu, Ariane Middel, **Kangning Huang**, Lei Zhao, TC Chakraborty, Zhihua Wang, Shasha Wang, Jiufeng Li, Long Li, Fan Huang, Yingying Ji, Xuecao Li & Manchun Li. 2025. “Exacerbated heat stress induced by urban browning in the Global South.” *Nature Cities*. 2, 157–169.
12. Yangzi Che, Xuecao Li, Xiaoping Liu, Xiaocong Xu, **Kangning Huang**, Peng Zhu, Qian Shi, Yimin Chen, Qiusheng Wu, Jay H Arehart, Wenping Yuan, Xia Li. 2024. “Mapping of individual building heights reveals the large gap of urban-rural living spaces in the contiguous US.” *The Innovation Geoscience*. 2(2): 100069.
13. Wei, Hong, Bin Chen, **Kangning Huang**, Meng Gao, Bin Fan, Tao Zhang, Ying Tu, and Bing Xu. 2024. “Moderating AC Usage Can Reduce Thermal Disparity between Indoor and Outdoor Environments.” *Environmental Science & Technology*. 58(24), 10524–10535.
14. Chenxi Hu, Chi-Yung Tam, Xinwei Li, **Kangning Huang**, Chao Ren, Kwun Yip Fung, Ziqian Wang. 2023. “Mega-city development impact on hourly extreme rainfall over the South China Greater Bay Area.” *Urban Climate*. 48, 101389.
15. **Kangning Huang**, Jiye Leng, Yong Xu, Xinwei Li, Meng Cai, Ran Wang, Chao Ren. 2021. “Facilitating urban climate forecasts in rapidly urbanizing regions with land-use change modeling.” *Urban Climate*. 36, 100806.
16. **Kangning Huang**, Xuhui Lee, Brian Stone Jr., Jason Knievel, Michelle L. Bell, Karen C. Seto. 2021. “Persistent increases in nighttime heat stress from urban expansion despite heat island mitigation.” *Journal of Geophysical Research: Atmospheres*. 126(4), e2020JD033831.
17. Pak Shing Yeung, Jimmy Chi-Hung Fung, Chao Ren, Yong Xu, **Kangning Huang**, Jiye Leng, Michael Mau-Fong Wong. 2020. “Investigating future urbanization’s impact on local climate under different climate change scenarios in mega-urban regions.” *Atmosphere*. 11(7), 771.
18. Yimin Chen, Xia Li, **Kangning Huang**, Ming Luo, Minyi Gao. 2020. “High-Resolution Gridded Population Projections for China Under the Shared Socioeconomic Pathways.” *Earth’s Future*. 8(6), e2020EF001491.
19. Xiaoping Liu, Yinghuai Huang, Xiaocong Xu, Xuecao Li, Xia Li, Philippe Ciais, Peirong Lin, Kai Gong, Alan Ziegler, Anping Chen, Peng Gong, Jun Chen, Guohua Hu, Yimin Chen, Shaojian Wang,

- Qiusheng Wu, **Kangning Huang**, Lyndon Estes, Zhenzhong Zeng. 2020. “High spatiotemporal resolution mapping of global urban change from 1985 to 2015.” *Nature Sustainability*. 3, 564–570.
20. Guangzhao Chen, Xia Li, Xiaoping Liu, Xun Liang, Jiye Leng, Yao Yao, Yue’an Qiu, Qianlian Wu, **Kangning Huang**. 2020. “Global projections of future urban land expansion under shared socioeconomic pathways.” *Nature Communications*. 11(1): 537.
 21. **Kangning Huang**, Xia Li, Xiaoping Liu, Karen C. Seto. 2019. “Projecting global urban land expansion and heat island intensification through 2050.” *Environmental Research Letters*. 14(11): 114037.
Media: [Yale News](#), [E&E News](#), [Scientific American](#); Data: [WRI Resource Watch](#)
 22. Tango Hu, Jiahong Liu, Gang Zheng, Dengrong Zhang, **Kangning Huang**. 2019. “Evaluation of historical and future wetland degradation using remote sensing imagery and land-use modeling.” *Land Degradation and Development*. 31(1): 65–80.
 23. Robert I. McDonald, Andressa V. Mansur, Fernando Ascensão, M’Lisa Colbert, Katie Crossman, Thomas Elmqvist, Andrew Gonzalez, Burak Güneralp, Dagmar Haase, Maike Hamann, Oliver Hillel, **Kangning Huang**, Belinda Kahnt, David Maddox, Andrea Pacheco, Henrique Pereira, Karen Seto, Rohan Simkin, Brenna Walsh, Alexandra S Werner, Carly Ziter. 2019. “Research gaps in knowledge of the impact of urban growth on biodiversity.” *Nature Sustainability*. 3: 16–24.
 24. Siyu Chen, Xiaorui Zhang, Jintai Lin, Jianping Huang, Dan Zhao, Tiangang Yuan, **Kangning Huang**, Yuan Luo, Zhuo Jia, Zhuo Zang, Yue’an Qiu, Li Xie. 2019. “Fugitive road dust PM_{2.5} emission and their potential health impacts.” *Environmental Science & Technology*. 53(14): 8455–8465.
 25. Xiaoping Liu, Shuli Chen, Li Zhuo, Jun Li, **Kangning Huang**. 2018. “Multi-sensor image registration by combining local self-similarity matching and mutual information.” *Frontiers of Earth Science*. 12(4): 779–790.
 26. Siyu Chen, Nanxuan Jiang, Jianping Huang, Xiaoguang Xu, Huiwei Zhang, Zhou Zhang, **Kangning Huang**, Xiaocong Xu, Yun Wei, Xiaodan Guan, Xiaorui Zhang, Yuan Luo, Zhiyuan Hu, Taichen Feng. 2018. “Quantifying contributions of natural and anthropogenic dust emission from different climate regions.” *Atmospheric Environment*. 191: 94–104.
 27. Jiayong Liang, Xiaoping Liu, **Kangning Huang**, Xia Li, Xun Shi, Yaning Chen. 2015. “Improved snow depth retrieval by integrating microwave brightness temperature and visible/infrared reflectance.” *Remote Sensing of Environment*. 156: 500–509.
 28. Jiayong Liang, Xiaoping Liu, **Kangning Huang**, Xia Li, Dagang Wang, Xianwei Wang. 2013. “Automatic registration of multi-sensor images using an integrated spatial and mutual information (SMI) metric.” *IEEE Transactions on Geoscience and Remote Sensing*. 52(1): 603–615.
 29. **Kangning Huang**, Xiaoping Liu, Xia Li, Jiayong Liang, Shenjing He. 2012. “An improved artificial immune system for seeking the Pareto front of land-use allocation problem in large areas.” *International Journal of Geographical Information Science*. 27(5): 922–946.
 30. Xiaoping Liu, Xia Li, Xun Shi, **Kangning Huang**, Yilun Liu. 2012. “A multi-type ant colony optimization (MACO) method for optimal land use allocation in large areas.” *International Journal of Geographical Information Science*. 26(7): 1325–1343.

Non-Peer-Reviewed Articles

1. **Kangning Huang**. 2022. “Urban forests face climate risks.” *Nature Climate Change | News & Views*. 12(10), 893–894.

Books

1. Kelly Aho, Tirthankar Chakraborty, Bowen Fang, **Kangning Huang**, Ava Liang, Natalie Schultz, Charlotte Stanley, Anna Walsh, Zhongwang Wei, Yichen Yang, Bowen Zhao, Xuhui Lee. 2017. *Fundamentals of Boundary-Layer Meteorology: Solutions Manual*. Springer International Publishing.

ISBN: 978-3-319-60853-2.

Reports

1. Lead author. *Nature in the Urban Century: A global assessment of where and how to conserve nature for biodiversity and human wellbeing*. 2018. [The Nature Conservancy](#).

PRESENTATIONS AND CONFERENCES

Invited Talks

- 2025 “AI for Urban Sustainability: Understanding, Simulating, and Managing Cities.” ADIA Lab “Sustainable AI” Summer School, Fields Institute, Shanghai.
- 2025 “AI for Urban Sustainability.” Urban Science and Sustainability Summer School, CAS Institute of Urban Environment, Xiamen.
- 2024 “Which Chinese Cities are Overbuilt? Insights from Urban Complexity Science.” China Crossroads, Shanghai; The Chinese University of Hong Kong.
- 2024 “From Sources of Prosperity to Potential Challenges: The Influences of Rivers and Seas for Shanghai and New York.” Conference on Common Ideals and Common Future, Shanghai.
- 2023 “The applications of spatial analysis in urban and climate health research.” Chinese Preventive Medicine Association, Qionghai, Hainan.
- 2023 “Climate trade-offs in urbanization in China and beyond.” China Crossroads, Shanghai.
- 2019 “Future global urban land expansion and limits of urban adaptations.” Hotspot Cities Symposium, University of Pennsylvania.
- 2017 “Forecasting urban land expansion and heat island intensification globally through 2050.” Silk Road Innovation Forum, Xi’an, China.
- 2017 “Urbanization, climate adaptation and resource constraints.” The Hong Kong Polytechnic University.

Sessions Organized

- 2018 “Building resilience to natural hazards by utilizing citizen science and understanding compound hazards.” AGU Fall Meeting, Washington, DC.

Selected Oral Presentations

- 2026 “Urban climbing plants as climate adaptation infrastructure: a global meta-analysis of cooling performance and deployment potential.” Association for Tropical Biology and Conservation (ATBC), Xishuangbanna, China.
- 2025 “Sponge Cities After a Decade.” Asia Oceania Geosciences Society (AOGS), Singapore.
- 2025 “Declining urban density attenuates rising population exposure to surface heat extremes.” International Conference of Geoinformatics, Jiaozhuo, China.
- 2024 “Mapping the mismatch: The urbanization of the built environment versus people in modern China.” AGU Fall Meeting, Washington, DC.
- 2024 “The urban mosaic of heat exposure.” IEEE IGARSS, Athens, Greece.
- 2023 “Taller or denser buildings? Effects of urban forms on heat island extremes.” AGU Fall Meeting, San Francisco.
- 2019 “Urban expansion neutralizes heatwave-risk reduction by socioeconomic development.” 16th International Conference on Urban Health, Xiamen.
- 2018 “Escalating heat stress by mid-century from large-scale urban land expansion.” AGU Fall Meeting, Washington, DC.
- 2018 “Global forecasts of urban land expansion and heat island intensification by 2050.” International Conference on Upscaling Urban Solutions, Berlin.
- 2018 “A spatial lag model for global urban heat island.” AAG Annual Meeting, New Orleans.

- 2016 “Contributions to augmented compound urban temperature extreme (ACUTE).” Urban Transitions Global Summit, Shanghai.
- 2013 “Improving un-gauged hydrological modeling by assimilating GRACE Terrestrial Water Storage data.” AGU Fall Meeting, San Francisco.
- 2012 “An artificial immune system for multi-objective land-use allocation.” International Conference on GeoInformatics, Hong Kong.

TEACHING EXPERIENCE

- 2022–2025** **Instructor**, “Environmental System Science,” [SOCS-SHU 204](#), NYU Shanghai.
- 2022–2025** **Instructor**, “Cities at a Crossroads: Environmental Challenges and Opportunities in Cities,” [SOCS-SHU 208](#), NYU Shanghai.
- 2022** **Guest lecturer**, “Climate and Life,” ENVST-UA 385, NYU New York.
- 2021** **Guest lecturer**, “Environmental Studies Senior Seminar,” ENVST-UA 900, NYU New York.
- 2020** **Guest lecturer**, “Natural and Socioeconomic Factors Shaping Environmental Health,” [One Health Training](#), Animal Dialogue.
- 2020** **Instructor**, “Introduction to Google Earth Engine,” [Research Application Laboratory](#), NCAR.
- 2015–2017** **Teaching Assistant**, “Modeling Geographic Objects,” Yale School of Forestry and Environmental Studies. Instructor: C. Dana Tomlin.

SERVICE

Journal Manuscript Review

Environmental Research Letters; Sustainable Cities and Society; International Journal of Geographical Information Science; Nature Geoscience; Nature Climate Change; Nature Cities; Journal of Geophysical Research—Atmospheres; npj Urban Sustainability; npj Environmental Social Science; MDPI: Land; Urban Climate; Frontiers of Earth Science.

Journal Editorship

Guest editor, [Atmosphere](#), Special Issue “Hazards, Urbanization, and Climate Change.”

Report Review

- *Heat Wave Guide for Cities*. 2019. Red Cross Red Crescent Climate Centre.
- *Guide for Urban Integrated Hydro-Meteorological, Climate and Environmental Services*. 2018. World Meteorology Organization.

Departmental Service

- Faculty Lunchtime Speaker Seminar Organizer, NYU Shanghai.
- Program Assessment Committee, NYU Shanghai, Area of Social Science.
- Faculty Continuous Contract Hiring Committee, NYU Shanghai, Area of Social Science.
- Faculty Continuous Contract Renewal Committee, NYU Shanghai, Area of Social Science.
- Developer of [Cambodian Genocide Program Geographic Database](#), Yale University Library.
- Member of [Professional Development Committee](#), NCAR Advanced Study Program.

MEDIA COVERAGE

- “Study: Swath of Natural Habitat Larger Than the UK Will Be Urbanized By 2030.” *Texas A&M Today*, 2020-01-30.
- “Urban Heat Islands Mean Warming Will Be Worse in Cities.” *Scientific American* / *E&E News*,

2019-11-21.

- “For Some Urban Areas, Warming Climate is Only Half the Threat.” *Yale News / Science Daily*, 2019-11-14.
- “Respite from the Urban Heat Island? Study Eyes Costs of Irrigation as Remedy.” *Yale News*, 2017-07-20.

PROFESSIONAL SKILLS

Land Use / Cover Change Modeling; Hydrometeorological Modeling; Spatial Data Analysis; Nonlinear Optimization; Data Assimilation; Image Processing.

Software: ArcGIS, QGIS, Google Earth Engine, ENVI, R, MATLAB, NCL, Fortran, Python.

LANGUAGES

English (fluent); Mandarin (native); Cantonese (native).

PROFESSIONAL MEMBERSHIP

American Geophysical Union.